



Sources of Joint Gains in Negotiations

Individuals enter into an agreement only if each party is better off with that agreement than without it. Put differently, the agreement represents *joint gains* to each party relative to her *Best Alternative To Negotiated Agreement* (BATNA). To find these joint gains when they exist, a negotiator must possess a keen understanding of how joint gains are created. In this note, we discuss several potential sources of joint gains. At one level, these sources can be viewed as a laundry list of joint gains possibility. However useful such a list may be, we find it more insightful to view this list through two organizing principles:

Principle 1: Most joint gains are about differences.

Principle 2: Joint gains are created when all parties receive something valuable for them in exchange for giving something cheap to them.

It is often said that “differences divide.” In negotiation, differences often enable agreement. To see this, imagine a world in which everybody is exactly the same: same tastes, desires, opinions, values, attitudes toward risk, etc. In this world, precisely because everything is worth exactly as much to me as it is to you, there is little opportunity for exchange. Now let’s return to our real world, a world of heterogeneity. In this note, we use these differences as the basic raw material and develop recipes based on Principles 1 and 2 that produce joint gains or “create value.”¹ We use simple examples to illustrate the basic analytical principles behind creation of these joint gains.

Potential Sources of Joint Gains

Differences in Interests

In their best-selling book, *Getting To Yes*, Fisher and Ury advise negotiators to “Focus on Interests, Not Positions.” Perhaps the most common way of creating joint gains is by exploiting differences in interests. In a negotiation, interests manifest themselves in terms of issues, particular means of satisfying more fundamental interests. For example, union and management may be

¹ Note that Principle 1 reads that “*most* joint gains are about differences.” There are categories of joint gains that do not involve differences. Sometimes there is no conflict on a particular issue: *all* parties want the same resolution of that issue. In this case, negotiators have certain “shared interests,” e.g., in the reputation of the company, the organizational structure of a firm, or in the relationship between the negotiating parties. In other cases, identical negotiators can achieve mutually beneficial economies of scale. For a further elaboration of this ideas, see David A. Lax and James K. Sebenius (1986), *The Manager as Negotiator*, New York: Free Press, 106-111.

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bargaining over the issue of work rules, but the underlying interests might be autonomy, personal satisfaction, etc. Management may well be able to meet their interests through changes other than changes in formal work rules.

Bob may put more weight on interest X than interest Y, while Carol cares more about interest Y than interest X. This combination of interests suggests an agreement in which Bob “gets his way” on interest X, while Carol “gets her way” on interest Y. Both would be better off than with a 50-50 X-Y split. The following example further illuminates these basic ideas:

Two small business partners, Danielle and Evelyn, are equally adept at accounting and sales. Danielle enjoys both accounting and sales, but prefers working with numbers to meeting with customers. Her partner, Evelyn, is indifferent to doing the accounting and taking responsibility for sales.

It is quite easy to imagine Danielle and Evelyn splitting the accounting and sales duties evenly. However, common sense dictates that Danielle should do the books, leaving Evelyn with the sales function. Indeed, both Danielle and Evelyn are happier with such an arrangement than with splitting the tasks evenly.

To make these concepts clearer, imagine that Danielle and Evelyn have scoring systems. Scoring systems are merely a convenient way of representing Danielle and Evelyn’s preferences for one arrangement over another.² If Danielle prefers arrangement X to arrangement Y, then Danielle should assign arrangement X more points than arrangement Y.

We might envision three possible allocations of effort: Danielle does all the sales and Evelyn does all of the accounting (Allocation A); Danielle and Evelyn split the accounting and sales evenly (Allocation B), and Danielle does all of the accounting and Evelyn does all of the sales (Allocation C).

Allocation	Danielle Effort Allocation		Value to Danielle			Evelyn Effort Allocation		Value to Evelyn		
	Acct’ng	Sales	Acct’ng	Sales	Total	Acct’ng	Sales	Acct’ng	Sales	Total
A	0%	100%	0	4	4	100%	0%	10	0	10
B	50%	50%	5	2	7	50%	50%	5	5	10
C	100%	0%	10	0	10	0%	100%	0	10	10

The scoring system reflects Danielle’s preferences for accounting over sales. She assigns 10 points to spending 100% of her time doing accounting and only 4 points to working full time on sales. The scoring system also reflects Evelyn’s indifference to accounting and sales: Evelyn assigns 10 points both to spending all her time doing accounting and all her time doing sales.

If Danielle and Evelyn fail to explore their interests, each might end up working half-time on accounting and half-time on sales. A 50-50 split is worth 7 points to Danielle, and 10 points to Evelyn. Is there a better arrangement? Sure. Since Evelyn doesn’t care whether she does accounting

² We do not go into detail on how to create these scoring systems. Interested readers may wish to reference Ralph L. Keeney and Howard Raiffa (1992), “Structuring and Analyzing Values for Multiple-Issue Negotiations,” in *Negotiation Analysis*, ed. Peyton Young, Ann Arbor: University of Michigan Press, 131-151, or Howard Raiffa and James K. Sebenius, “Analytics for Negotiators: Seven Dialogs,” Dialog 5.

or sales, and Danielle strongly prefers to do the numbers, a much better division of labor is to have Danielle do all the accounting, leaving Evelyn to carry out the sales work. Such a division of work leaves Danielle and Evelyn each with 10 points, clearly better than if the activities had been split evenly.

Couples, business partners, organizations, and nations often craft agreements that exploit differences in how the parties value the interests. In a bilateral negotiation, the parties may be diametrically opposed on all issues: the best resolution of an issue for Party *A* is the worst resolution of that issue for Party *B*, and vice versa. Still, there may be ample opportunity for an agreement. There is almost always some issue *X* that is less important to *A* than it is to *B*, and consequently another issue *Y* that is paramount for *A* and insignificant for *B*. Joint gains are created if issue *X* is resolved in *B*'s favor, whereas issue *Y* is resolved in *A*'s favor. Recall Principle 2: *Joint gains are created when all parties receive something valuable for them in exchange for something cheap for them.*

Recipe (differences in interests): *Concede on an issue that is relatively unimportant to you (and relatively important to your counterpart) in exchange for a concession on an issue that you consider more important (and is relatively unimportant to your counterpart).*

Differences in Opinion

Mark Twain wrote: "It were not best that we should all think alike; it is difference of opinion that makes horse races." Individuals often differ in beliefs about what the future holds in store, whether those differences concern the future unit sales of a product, the price of oil, the Yen-U.S. dollar exchange rate, the likelihood of a technological break-through, or the productivity of a critical research scientist. Divergent views are often implicit in the sale of a stock or a company from one individual to another: the seller believes that the stock price will drop, while the buyer believes that an increase in the price of the stock is imminent.

There are a number of reasons that individuals may have different beliefs about the likelihood of a critical event, including varying prior exposure to information or differences in how available information is interpreted.³ Whatever the reason for such differences, disagreements of this sort are raw material for joint gains in the form of *contingent agreements*, agreements in which the payments from one party to another party are *contingent* on a particular event. Let's turn to an example that illustrates this basic idea:

Franklin Publishers is negotiating with Gloria Gill on rights for Gloria's latest novel. The two parties are at an impasse, largely because Gloria believes the book will be a best seller, whereas Franklin's forecasts are more modest. Furthermore, neither side has been successful in convincing the other that their estimates are mistaken.

Franklin and Gloria may well fail to reach an agreement, each trying desperately to narrow the differences: "Don't you see when you have a great opportunity? This book is destined to be a best seller, if I do say so myself." "You're far too optimistic. It's just like *Great Hopes*, a novel which you might remember sold a puny 10,000 copies." And so on. There's an alternative: Franklin and Gloria can allow the disagreements to stand as they are, differences about what the future holds in store. In turn, they can exploit these differences by constructing a contingent agreement.

³ For an account of how probabilistic judgments are sensitive to informational cues, see Amos Tversky and Daniel Kahneman (1974), "Judgment under uncertainty: Heuristics and biases," *Science* 185, 1124-1131.

Suppose for simplicity that Gloria's book will sell either 50,000 (moderate seller) or 100,000 (best seller) copies. Franklin believes that the probability of selling 50,000 books is .70, whereas the probability of Gloria's book becoming a best seller is .30. Gloria, of course, disagrees: she believes that it is more likely to be a best seller than a moderate seller. She assesses the probabilities of sales of 50,000 and 100,000 copies as .20 and .80 respectively.

To illustrate a contingent deal, suppose that Gloria needs to get a flat advance of \$140,000 for her novel, or else she will shop her book elsewhere. The most Franklin can offer is \$120,000. Since Gloria wants \$140,000, and the highest Franklin is prepared to offer is \$120,000, there is no possibility of an agreement involving *unconditional* payments. There is, however, a deal involving *conditional* payments. Imagine that Franklin offers to pay Gloria \$100,000, plus an additional \$60,000 if the book sells 100,000 copies. The expected value to Franklin is now, $.70 (\$100,000) + (.30) (\$160,000) = \$108,000$, better than Franklin's maximum. For Gloria, the expected value of the new arrangement is $.20 (\$100,000) + (.80) (\$160,000) = \$148,000$, which exceeds her minimum. There's a deal!

	Number of Copies Sold		Expected Value
	50,000 copies	100,000 copies	
Franklin's payment to Gloria	\$100,000	\$160,000	
Cost to Franklin	.70	.30	\$118,000
Benefit to Gloria	.20	.80	\$148,000

What has happened? Returning to Principle 2, we see that Franklin and Gloria each have exchanged something expected to be cheap to one for something the other expects to be valuable. Since Franklin believes that the chance of selling 100,000 copies is only .30, it is not terribly costly for him to offer Gloria a higher sum in that event. In return, Franklin pays less, \$100,000 if the book only sells 50,000 copies, an event that Franklin believes is quite likely. Gloria, on the other hand, is willing to take a lower minimum payment, \$100,000, because she believes that there is .80 chance that she will actually receive an additional \$60,000. In other words, each has exchanged something cheap to her in expected value terms for something she judges to be valuable in expected value terms. Once again, we have an application of Principle 2.

Franklin and Gloria can both do even better by lowering the minimum payment, and increasing Franklin's payment to Gloria in the event that Gloria's book is a best seller. The following payment structure has a higher expected value for both Franklin and Gloria.

	Number of Copies Sold		Expected Value
	50,000 copies	100,000 copies	
Franklin's payment to Gloria	\$50,000	\$200,000	
Cost to Franklin	.70	.30	\$95,000
Benefit to Gloria	.20	.80	\$170,000

In fact, mathematically, there are no limits to the expected value joint gains that can be created. In the following example, Gloria pays Franklin \$5,000,000 in the event that her book sells 50,000 copies, but receives \$5,000,000 if her book becomes a best seller. Both Franklin and Gloria *make* money in expected value terms when the deal is structured in this fashion!

	Number of Copies Sold		Expected Value
	50,000 copies	100,000 copies	
Franklin's payment to Gloria	\$-5,000,000	\$5,000,000	
<i>Benefit</i> to Franklin	.70	.30	\$2,000,000
Benefit to Gloria	.20	.80	\$3,000,000

Why does this arrangement seem so contrived and unrealistic? First, agreements must meet the test of *institutional plausibility*. In this case, it seems so crazy for *Gloria to pay Franklin \$5,000,000* that we hope that neither party would partake of such an agreement, whatever the magnitude of expected value gains. It is important to note that the efficiency of the scheme stems from betting, which could occur outside the setting of this deal. The deal (or perhaps lack of an agreement in the absence of betting) gives both parties a legitimate reason to bet.

A second problem with the last example is risk aversion. Gloria and Franklin might well be willing to make decisions on the basis of expected value for small stakes and even for several hundred thousand dollars. But few would consider \$5,000,000 small stakes. Most individuals and even most firms would think many times about risking that large a sum of money. In the next section, we talk explicitly about risk aversion and how differences in risk attitudes can be turned into joint gains.

Finally, some might object that these gains aren't real in the sense that it will turn out that one party is "right" and one party is "wrong." Put differently, some question whether contingent contracts actually do create any economic value. To argue that such transactions *do* have economic value, we turn to the insurance business. Insurance, too, has a winner or a loser. Those who are unfortunate enough to suffer a fire are "winners" in the sense of having made the "right" decision to insure their house, while those whose house is not inflicted by fire made the "wrong" decision to purchase the policy. Someone was "right" and someone was "wrong" at the end of the policy's term. Yet, it seems difficult to argue that insurance provides no economic benefit. Similarly, although contingent agreements have a "winner" and a "loser" at the end, the contract serves an economic purpose: it has allowed a transaction to take place that might not have taken place otherwise.

The following recipe summarizes this section:

Recipe (differences in opinion): *Create a contingent arrangement or "bet." Structure the agreement as follows:*

If A believes that some event is less likely than B does, then A should pay B if that event occurs (perhaps "compensating" A with something of greater value to A).

Differences in Risk Preference

In the previous section, we assumed that everyone maximized expected value. However, we know that people often pay to avoid risk, whether by purchasing insurance or by sacrificing some potential "upside" for a bit more security. Naturally, there are substantial individual differences in the tendency to tolerate risk. Such differences can be exploited to create joint gains. We turn to the following example:

Two business partners, Hank and Ida, own equal shares in a venture that has a 50% chance of becoming a bonanza. If the venture goes bonanza, Hank and Ida will be rewarded with \$1,000,000 in profits. If the venture fails to go bonanza, the venture will just break even, producing no profit.

Hank and Ida face a problem of how to divide the future proceeds of the venture. Suppose that unlike the book negotiation, there are not differences in opinion to exploit, only differences in willingness to tolerate risk. Hank is extremely risk averse, cautious in his personal and business finances. Ida, on the other hand, is glad to make decisions, even large stakes ones, on the basis of expected value.

Suppose it is proposed that Hank and Ida each get half of the proceeds, in all cases. In other words, Hank and Ida will each receive \$500,000 if the venture goes bonanza, and nothing otherwise. Is there a division of proceeds that would make both Hank and Ida happier? Yes, one that exploits differences in risk preferences. Since Hank is risk-averse, suppose that he would be glad to take \$100,000 for sure over the proposed alternative, a .50 chance at \$500,000 and a .50 chance at \$0. What's left over goes to Ida: she receives \$900,000 if the venture goes bonanza, and pays Hank \$100,000 in the event that the venture fails. Since Ida is willing to make decisions on the basis of expected value, this arrangement is worth \$400,000 to her, substantially higher than the \$250,000 expected value of the previous arrangement. Hank was glad to trade the "upside potential" for some security. Again we see an application of Principle 2: Hank traded "something cheap" for "something valuable." In turn, Ida assumes substantial additional risk but is rewarded with the possibility of a large potential payment.

	<u>Venture Goes Bonanza</u>	<u>Venture Fails</u>	
	<u>.50</u>	<u>.50</u>	<u>Value</u>
Hank's payment	\$100,000	\$100,000	\$100,000
Ida's payment	\$900,000	\$-100,000	\$400,000
Total payment	\$1,000,000	\$0	

To sum up this section:

Recipe (differences in risk preference): *Transfer risk to the party who is more willing to bear the risk (and compensate elsewhere).*

Differences in Time Preference

Individuals or firms may have substantially different discount rates, a measure of how they trade off consumption or money taken now with consumption or money taken in the future. Firms, for example, may differ in terms of the riskiness of the projects that they are currently undertaking, as well as their access to capital markets or other sources of funds. Individuals, too, may differ in their patience, liquidity, or their career stage. In this section, we illustrate how differences in time preference can be leveraged for the creation of joint gains. We use the following example:

Jersey Tool Company and Kennington Products have just agreed on a five-year contract, in which Jersey will sell Kennington \$10,000 in tools each year for the next five years.

A natural arrangement would involve annual payments of \$10,000 from Kennington to Jersey. Let's suppose that Jersey has a 25% annual discount rate. On the other hand, Kennington has a considerably lower discount rate of 15%. Are there any ways of restructuring the payment schedule to reflect these differences in time preference?

Let's first look at expediting the \$10,000 payment in Year 1. To Jersey, \$10,000 in Year 1 is worth $\$10,000 / 1.25 = \$8,000$ to Jersey now. Any current payment over \$8,000 is better for Jersey than \$10,000 in Year 1. On the other hand, the present value of a payment of \$10,000 in year 1 is $\$10,000 / 1.15 = \$8,696$. Thus, Kennington would be happy to pay as much \$8,696 today to forego a \$10,000 payment in Year 1. Putting the two pieces together, Jersey and Kennington are better off if Kennington pays Jersey anything between \$8,000 and \$8,696 today.

Again, the value created by expediting payments is based on Principle 2. Expediting payments benefits Jersey more than it hurts Kennington. Of course, the general solution does not always involve expediting payments. If Jersey's discount rate had been 10% instead of 25%, then value would be created by *delaying* payments. The general idea is that when time differences exist, payments or benefits should be shifted (either moved up or pushed back in time) to reflect these differences.

Recipe (differences in time preference): *Shift the payments or benefits to reflect differences in discount rates (and compensate elsewhere).*

Other Differences

There are a number of other types of differences that can be capitalized upon. Different tax rates suggest shifting income toward the party with the lower tax rate. Different and complementary capabilities suggest a partnership or alliance in which all partners are better off. If there are differences in motivation, perhaps deal should be structured to provide the less motivated individual with incentive payments. If one party wants a public "win" and the other parties care mainly about substance, create a press conference to announce victory for one side, leaving the "fine print" to the others. In all cases, the creation of joint gains is based on Principle 2.

Summary

We have discussed several sources of joint gains, as well as presenting some basic principles that organize this list and can be used for identifying other sources of gains. In practice, finding and exploiting joint gains is not a matter of luck or even simply a matter of skill—thorough preparation is needed as well. Below we outline some steps in the preparation stage.

First, construct a *differences inventory*. Remember that differences, rather than common ground, are the raw material for most joint gains. Start by identifying and understanding your interests, your attitudes toward risk, your discount rate, as well as your beliefs about critical uncertainties. As difficult as this task might be, you must also do the same for your negotiating counterparts. Probe as much as possible, but don't be afraid to admit that you simply don't know. This might indicate an opportunity for collecting information during the negotiation.

Second, *envision agreements*. What are the critical differences? What might deals based on these differences look like? Are there barriers to agreeing upon these deals? Coming to the negotiating table with an understanding of how to create gains allows you to control the negotiation process, create joint gains, as well as possibly grab the lion's share of the gains for yourself.

Third, *identify what information you need*. What's missing from the differences inventory? You may very well not know how your counterpart feels about the price of oil or the interest rate. You might not know whether she values autonomy or compensation more. Your counterpart's cost of capital might be a complete mystery to you. The nature of the agreement you strike depends on the answers to these questions. It should be obvious how important it is to learn about your counterpart. When information is needed for crafting good agreements, it is critical to manage the process, fostering information sharing, and managing the tendency to escalate conflict. You must also be aware of the potential dangers of misrepresentation, yours or theirs. For example, if one side misrepresents its discount rate, you might end up with a deal in which payments are expedited, when an efficient agreement calls for delaying payments.